Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18EE31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

ELECTRIC CIRCUIT ANALYSIS

Time: 3 Hours **Maximum Marks:70**

Instructions: i. Answer **FIVE** full Questions.

ii. Questions 1 and 2 are compulsory.

1. a. Find the voltage Vx in the network shown in fig.1a. Use mesh method. **06**

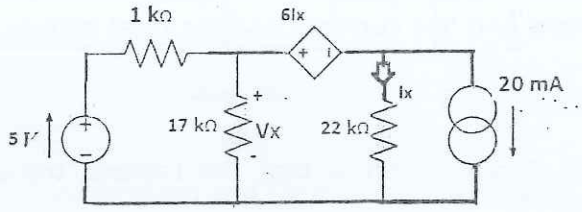


Fig.1a.

b. For the network shown in fig. 1b, find the value of R_B for maximum power transfer through it and also find the maximum power. **08**

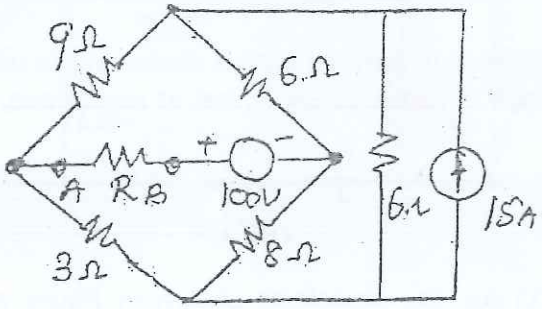


Fig.1b.

2. a. Explain the behavior of R, L and C at t=0+ and at t=infinity with and without initial stored energy. **05**

b. An initially relaxed series R-L-C circuit of value R=10 Ω, L=1 H and C=100 μF is connected across a DC source of 100 V by closing a switch at t=0. Find the initial value at t=0+ and its first and second derivatives at t=0+. **05**

c. Define the time constant of R-L and R-C circuits. What is its significance? **04**

3. a. Two identical coils when connected in series across a source, the total inductance is found to be 3.0 H. When one of the coils is reversed the total inductance become 1.4 H. Find the mutual inductance between them and the self-inductance of each coil. Also find the coupling coefficient. **06**

b. A star connected impedance draws a power of 50 kW and a leading current of 100 A from a three phase 415 V, 50 Hz, R,Y,B phase sequence balanced supply. Find the impedance and the circuit elements. Also find the reactive power and the power factor. What will happen to these results if the phase sequence is reversed? **08**

OR

4. a. What is a coupled circuit and dot convention and what is its importance? Briefly explain. **03**

b. A three phase unbalanced star connected load is excited by a balanced three phase supply. Derive an expression for the potential difference between the supply neutral and the load neutrals. Write expressions for unbalanced phase voltages and also draw the vector diagram. **05**

c. An unbalanced delta connected load of Z1=(10-j10) Ω, Z2=(20-j20) Ω and Z3=(25+j25) Ω are connected across a balanced three phase 415 V, 50 Hz supply. Find the phase and line currents and the total active, power drawn by this load. **06**

5. a. What is an impulse function? Show that the Laplace transformation of a unit impulse function is unity. **04**

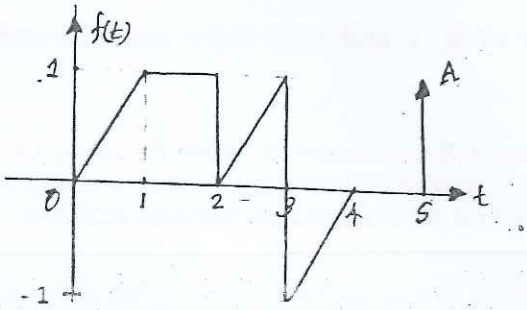
b. An R-L series circuit is energized by a supply of 10 V DC by closing a switch in series with it. R=10 Ω and L=5 mH. The circuit was under steady state before closing the switch. Find the current for time't' greater than zero. What is the current after 5 milli seconds? Use Laplace transformations. **06**

c. A fixed condenser is placed in parallel with a choke coil in which the resistance is fixed and the inductance is variable. Show that at resonance, **04**

$$X_L = \frac{X_C}{2} \pm \sqrt{\{(\frac{X_C}{2})^2 - R^2\}}$$

OR

6. a. Write the function f(t) for the waveform shown in figure and find the Laplace transformation. **05**



b. State and explain the initial and final value theorem and also mention its significance in the electrical circuits. **04**

c. An R-L-C series circuit has a capacitance of 320 pico Farad connected across a source of0.85 V has Q of 50 and resonant frequency of 175 kHz. Find the inductance, resistance, the current and the bandwidth of the circuit. **05**

7. a. Define z and y parameters and obtain the relationship between them. Draw the equivalent circuit of Z-parameters. **06**

b. Find the h and transmission parameters of the network shown in figure 7b. **08**

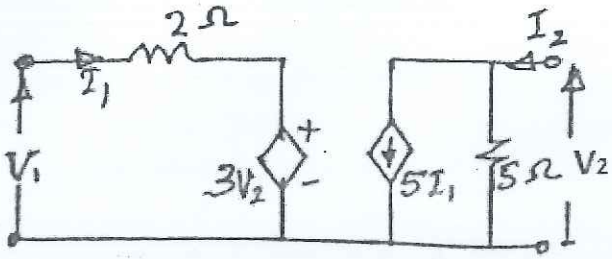


Fig 7b.

OR

8. a. Define h and transmission parameters and obtain the relationship between h and y parameters. **06**

b. Find the y and z parameters of the circuit shown in figure 8b. **08**

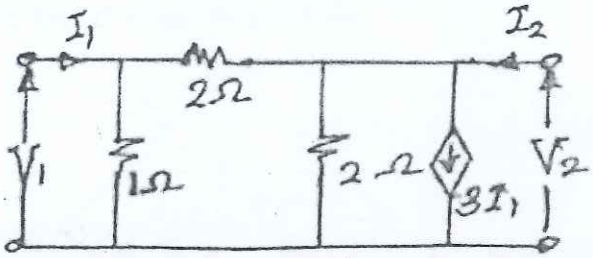


Fig.8b